Week 13 Individual Project Report Guidelines (Stream 1)

MATH3888 (Interdisciplinary maths project unit)

Submission Deadline: **Friday, November 6th, 23:59**.

Submission and Formatting Instructions

You will create your final individual report using the LaTeX typesetting system. You are encouraged to use Overleaf to create your LaTeX report which you can access via your browser through your University of Sydney account.

Submission of the corresponding PDF file is via Canvas.

- **pdf**-file of your individual project report.
- file name should be `groupXX-yourSID.pdf` where `XX` is your two-digit group number (01-15) and `yourSID` is your nine-digit Student ID number (e.g., `group07-412356789.pdf`).

Use the following basic setup for your LaTeX file:

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath}
\begin{document}
...
\end{document}
```

This individual project report is worth 30% of your final mark.

Constraints

- **Originality & Distinctness:** While based on the same project, this report must be your own original work and distinct from the Group Report. Directly copy-pasting text, figures, or analyses from the Group Report is self-plagiarism and will not result in a passing mark. You must write your own narrative.
- The final submitted PDF document shall consist of no more than 8 pages including figures and tables, but **not** including references and appendices.
- Include only 'key' figures/tables/schematics in the main body of the report. Complex figures/tables or detailed methodologies can be reported in Appendices.
- **Font Size & Margins:** The font size is strictly 11 points and the margins of the document must be set to 1 inch (e.g., using the `geometry` package as instructed above).
- The `amsmath` package might be needed for the mathematical editing. Add any other packages, if needed.

Guidelines for the Structure of Your Report

Please use the following headings (you can add subheadings):

Title

Succinct description of your project (ideally should fit on a line or two at most).

1. Introduction

This should be a general introduction to the overall project covering the relevant literature (refrain from citing material you found on the web; restrict to standard text books and original literature in peer-reviewed journals).

Keep the biological background brief; the primary focus of this report is your mathematical and computational analysis. Once you have provided the background and stated the problem or question under study, explain the purpose of your study. Include at the end of your Introduction a clear and exact statement of your study aims.

2. Methods

This section provides the details of how you conducted your study. You should focus entirely on reporting the discipline-specific (maths) aspects of the research. This must include aspects of mathematical theory and numerical algorithms & computational implementation.

Distinction/High Distinction Requirement: To achieve higher marks, you must go beyond simply reproducing the group's baseline textbook models or just repeating what was covered in lectures. While you may incorporate mathematical theory from the lectures, reproducing lecture and group report material will earn at most a Pass/Credit, provided the incorporated material is relevant. You may also expand upon mathematical theory directly related to the project explored in your own independent study (e.g., two-parameter bifurcation mapping, identifying exact boundaries of oscillatory regimes, or novel extensions to higher dimensions).

Note: This is YOUR individual report so we encourage you to present what YOU think are the best strategies/methods (this can differ to what the group has decided).

3. Results

Simply state what you found, but do not interpret the results or discuss their implications. Again, you should focus heavily on reporting the discipline-specific (maths) aspects of the research.

Results should be presented in a logical order. In general, this will be in order of importance. Do not duplicate data among figures, tables, and text. (If relevant, include the results of statistical analyses in the text).

4. Discussion

This section should be about the interpretation of your results. Discuss strengths and limitations of the mathematical approach you took AND place your research into the context of the overall project (are there other computational approaches that, with hindsight, might have been better?). For higher level marks, also place your research into the context of the current body of research in your chosen field. If your findings are preliminary, suggest future mathematical studies that need to be carried out.

Again, this is YOUR individual report so we encourage you to present YOUR point of view (this can differ to what the group has decided).

5. Conclusions

Final paragraph summarising keys points from the report.

References

Ideally use BibTeX with the `natbib`-style.

Appendices (optional)

There is no word limit for this additional material but it should be relevant material—don't just dump anything you can find here and hope it counts towards something. Any material presented here needs to be clearly linked and address an issue raised in the main body of the report.

The 3Cs of Scientific Presentation

Part of your task is to determine how best to present the information and your scientific argument. Think of the 3Cs:

- **Considered:** Have you considered all aspects of the project?
- **Constructed:** Is your report well constructed (structured)?
  Have you used diagrams/figures/maths & computational tools appropriately to convey your information?
- **Concise:** Being concise brings clarity. Is your report accurate and easy to read?

Grading Rubric

Criteria	Excellent (HD)	Very Good (D)	Good (CR)	OK (P)	Poor (F)
Knowledge of discipline area	Commanding depth of mathematical and computational theory. Demonstrates very strong understanding of the project's specific mathematical implications.	Comprehensive depth of mathematical knowledge with a strong focus on the specific computational methods, with appreciation of broader implications.	Sound knowledge of the immediate mathematical and computational methods, with some appreciation of their broader significance.	Satisfactory knowledge of the immediate mathematical methods, with limited appreciation of the broader computational context.	Poor or limited knowledge of the mathematical discipline area, or an over-reliance on biological background over mathematical theory.
Mathematical Content & Originality	Exceptional independent mathematical exploration (e.g., advanced bifurcations, novel higher-dimensional extensions). Distinctly unique from, and significantly extends beyond, the group baseline and lecture material.	Strong independent mathematical effort. Clear evidence of analysis extending beyond the standard group report and basic lecture material. Highly distinct individual narrative.	Solid mathematical analysis. Primarily relies on accurately applying lecture theory and group baseline models, with some minor independent computational insights.	Satisfactory mathematical analysis. Relies heavily on reproducing the group's baseline models or standard lecture material without meaningful independent extension.	Inadequate mathematical content. Unacceptable duplication of the group report, or merely copy-pasting lecture proofs with no individual application.
Report style, tables, figures	Clear concise and logical report. All figures and tables have legends and captions that enable them to be interpreted without extensive reference to the text.	Reasonably concise and logical. Some figure legends and captions may lack sufficient detail.	Report overall satisfactory but some sections should be more concise/display better logic. Figure legends and captions may be poorly explained.	Report overall satisfactory but should be more concise/display better logic. Figure legends and captions may be poorly explained.	Inadequate level of detail and/or logic. Missing figure legends and captions.